

Assembly and Calibration Guide for the Scale

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1 Introduction

This document provides step-by-step instructions on how to 3D print the parts, assemble the electronics, and calibrate your scale.

2 Parts and Preparations

2.1 3D-Printed Parts

- Main enclosure
- Base plate
- Bracket
- Anti-slip mat

2.2 Electronics

- LilyGO T-touch bar OLED (microcontroller)
- HX711 (load cell amplifier)
- Load cell (sensor)
- Battery (with battery connector)
- GPIO wires (for connections between microcontroller and HX711)

2.3 Tools and Supplies

- Soldering iron and solder
- Wire cutters (to trim cables)
- Screwdriver
- Reference weight (for calibration)
- Computer with Arduino IDE and VisualStudio with PlatformIO
- Optional: Hot glue gun or suitable adhesive for attaching TPU anti-slip mat

3 3D-Printing the Parts

1. Print all main parts
2. (Optional) Print the anti-slip pad or mold

4 Preparing the Electronics

4.1 Trimming the GPIO Wires

- Cut the GPIO cables to a length shorter than 55 mm.

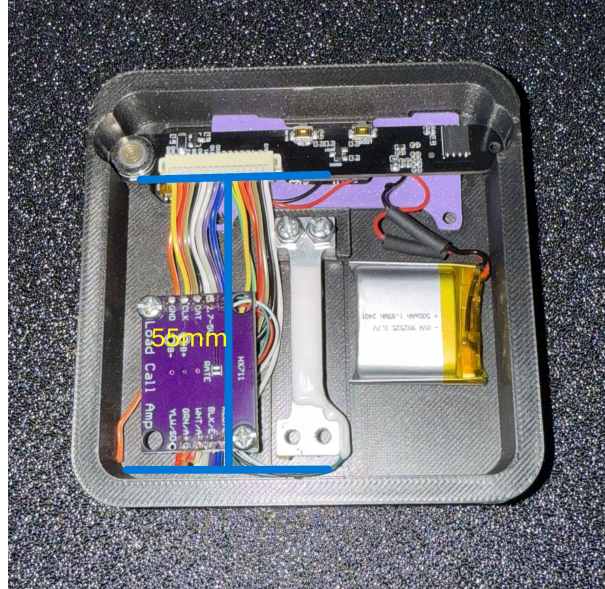


Figure 1: The length should be no longer than 55mm.

4.2 Soldering the Load Cell to the HX711

- Red to E+
- Black to E-
- White to A-
- Green to A+

4.3 Soldering ESP Pins to the HX711

- Included with the controller is a 16 wire (4) cable for gpio.
- Solder the following wires:

ESP Pin	HX711 Connection
4	DAT
3	CLK
6	VCC/2.7–5 V
GND	GND

Table 1: Connections from ESP to HX711



Figure 2: LilyGO Items included

4.4 Soldering the Battery Connector

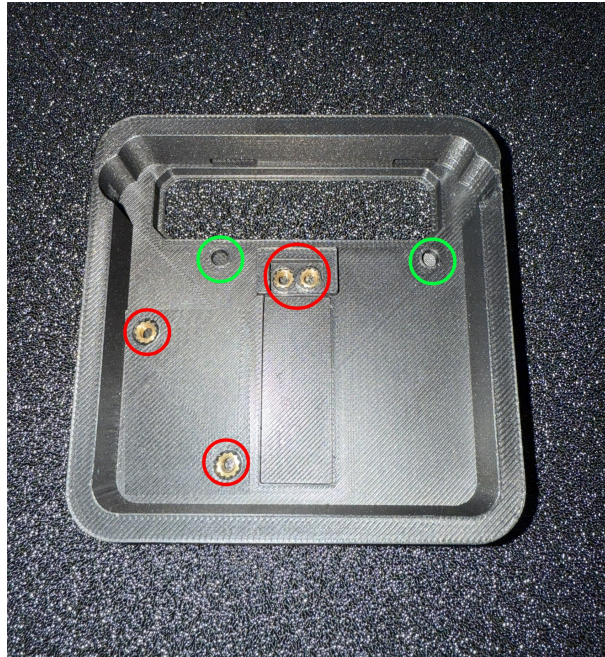
- Carefully solder the pigtail (2) included with the controller to the battery wires.

5 Test Everything

1. Connect the ESP to the HX711 using the wires you soldered in the previous step.
2. Plug the battery into the ESP.
3. Connect the device to a computer and check that it powers on correctly. (it should launch an example project)

6 Assembling the Unit

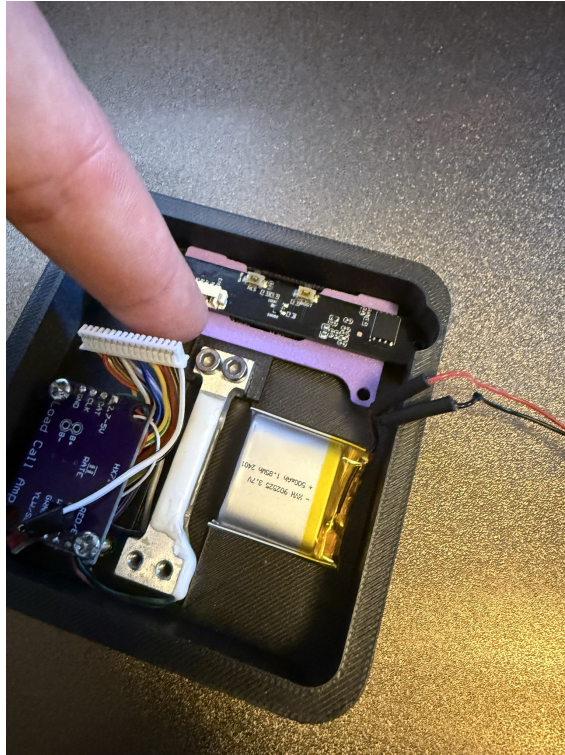
1. Insert all the threaded inserts as shown, please add in both red and green circles:



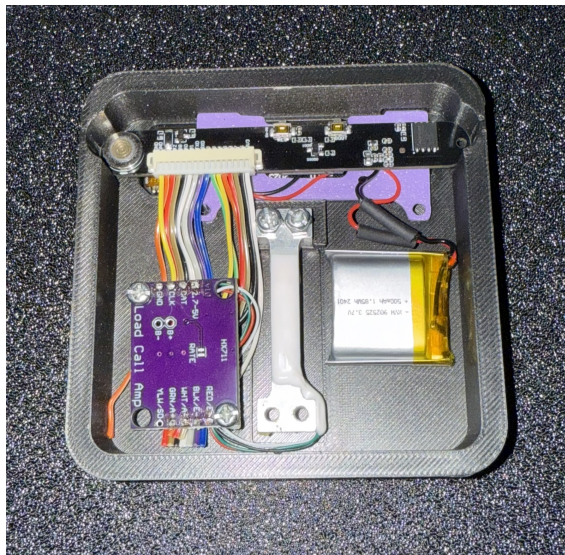
2. Put the bracket between the main board and display as shown:



3. Connect the battery cable and push the board down making sure to keep the board as flat as possible.
4. Push the bracket tabs into the slots in the enclosure and apply pressure down on the bracket as shown, it should click into place.



5. Screw the bracket down using the M3x6 screws
6. Now push the board down all the way until its securely resting on the bracket.
7. Place the hx711 into the main enclosure as shown routing the cables under it, screwing it down.



8. Make sure the load cell is mounted securely using the M3x10 screws. (NOTE: It should be attached with the non threaded side)
9. Mount the battery with double sided tape.
10. Attach the base plate using the M3x12 screws (Optional: use countersunk screws)

7 Calibrating the Scale

1. Obtain a reference weight (using another accurate scale or a dedicated calibration weight).
2. **Important:** Place the scale **upside down** for calibration.
3. Download hx711 library in arduinoIDE
4. Open the Arduino IDE and load the sketch named `Scale_calibration`. (Credits: SparkFun)
5. Upload the sketch to your ESP and open the Serial Monitor at 9600 baud.
6. When the weight readings appear, place the reference weight on the scale.
7. Use the + or - keys in the Serial Monitor to adjust the calibration factor until the displayed value matches the reference weight.
8. Note down the final calibration factor.

8 Final Firmware Installation

1. Follow the instructions on the project's GitHub repository to download the final firmware.
2. Open `scale.cpp` and insert the calibration factor you found.
3. Upload the final firmware to the ESP.
4. Verify that the scale now reads correctly when placing your reference weight.

8.1 OTA

To update go to "yourscaleIP"/update in any browser

1. Get the latest update from github and re-enter calibration factor
2. Generate a firmware.bin file by building program in platformio
3. Locate the file in `.pio/build/esp32s3box`
4. Drag and drop the file in PrettyOTA box
5. Wait for it to finish and reboot

9 Attaching the TPU Anti-Slip Mat (Optional)

1. Apply glue (e.g., hot glue) to the bottom of the scale.
2. Place the TPU mat carefully and press it in place.
3. Let the glue dry before using the scale.

10 Usage

- The battery powers the device; tells you when battery is low, recharge using provided cable.
- Touch left display to start/stop timer
- Touch right display to tare weight and reset time
- The scale goes into deep sleep when idle for 5 min.

- Touch any part of the display to wake up.
- For best accuracy, place the scale on a stable, flat surface.
- Avoid impact.

11 Troubleshooting

- **No readings:** Check if the HX711 is soldered correctly, and if the DAT/CLK lines are connected to the correct ESP pins.
- **Unstable readings:** Ensure wires are soldered well and that there are no shorts.
- **Constantly increasing weight:** Ensure that the base plate and the main enclosure can move independently, and that the load cell is free to "bend"
- **Incorrect calibration:** Repeat the calibration process with a more accurate reference weight. More accurate reference and calibration factor equals more accurate scale

12 Conclusion

Following these steps should result in a fully functional scale. Keep a record of your calibration factor in case you need to update or re-calibrate in the future. Good luck!

Contact / Support:

If you need additional help or guidance, please refer to the project's GitHub repository or contact me.